THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A solid metallic cylinder of base radius 6 cm and height 32 cm is melted and recast into solid spheres of radius 3 cm. The number of spheres obtained is: 1

(a) 32 **(b)** 38 **(c)** 64 **(d)** 30

2. The class mark of the class $10–25$ is: 1

(a) 7.5 **(b)** 17.5 **(c)** 35 **(d)** 15

3. A lot of 135 ball pens contains 15 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{119}{135}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{8}{9}$ **(d)** $\frac{121}{135}$

4. A solid metallic cone of base radius 6 cm and height 24 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

(a) 6 cm **(b)** 12 cm **(c)** 5 cm **(d)** 7 cm

5. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 8, 16, 8, 10, 8 respectively, The mean of the data is: 1

(a) 238 **(b)** 24.8 **(c)** 23.8 **(d)** 25

6. Two dice are thrown simultaneously. The probability that the sum of the numbers is 7 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{7}{36}$

7. A toy is a cone of base radius 21 cm and height 28 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 5082 cm² **(b)** 2310 cm² **(c)** 6468 cm² **(d)** 4620 cm²

8. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 2, 17, 12, 10, 9 respectively, The median of the data is: 1

(a) 60 **(b)** 70 **(c)** 66.4 **(d)** 65

9. A coin is tossed three times. The probability of getting at least two heads is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{4}{9}$

10. A capsule is a cylinder of radius 7 cm and length 14 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 1540 cm² **(b)** 2464 cm² **(c)** 924 cm² **(d)** 1232 cm²

11. For the data with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 3, 13, 5, 16, 7, 6 respectively, The mode of the data is: 1

(a) 80 **(b)** 90 **(c)** 80.8 **(d)** 85.5

12. A letter is chosen at random from the word 'PROBABILITY'. The probability that it is the letter B is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{9}{11}$ **(c)** $\frac{2}{11}$ **(d)** $\frac{1}{11}$

13. A solid is a cube of edge 28 cm surmounted by a hemisphere of diameter 28 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 4704 cm² **(b)** 5936 cm² **(c)** 4088 cm² **(d)** 5320 cm²

14. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 5, 4, 10, 4, 2 respectively, The modal class is: 1

(a) 20–30 **(b)** 10–20 **(c)** 30–40 **(d)** 40–50

15. A bag contains 5 red and 5 blue balls. 5 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{11}{15}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A red die and a blue die are rolled together. Find the probability that the sum is less than 5. 2

17. A solid sphere of radius 3 cm is completely dropped into water in a cylindrical vessel of base radius 3 cm, and no water spills. Find the rise in the water level. 2

18. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 2

Lifetime (hours)	0–10	10–20	20–30	30–40	40–50	50–60
Number of bulbs	16	22	20	5	19	18

Find the median.

19. A letter is chosen at random from the word 'EDUCATION'. Find the probability that it is a vowel. 2

20. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. Find the length of the wire. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The daily electricity consumption (in units) of households in a colony is given below. 3

Units consumed	20–30	30–40	40–50	50–60	60–70
Number of households	3	16	20	4	7

Find the mode.

22. A bag contains 3 red and 9 blue balls. 3 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 3

23. A toy is a cone of base radius 21 cm and height 20 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

24. The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	60–70	70–80	80–90	90–100	100–110
Number of workers	12	20	3	2	3

Find the mean.

OR

Two dice are thrown simultaneously. Find the probability that the sum is a multiple of 4. 3

25. A capsule is a cylinder of radius 10.5 cm and length 20 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

OR

The weights (in kg) of students of a class are given below. 3

Weight (kg)	10–20	20–30	30–40	40–50	50–60
Number of students	2	4	2	15	2

Find the median.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 15, 3, x, 16, y, 10, 13 respectively, and total frequency 100, is 36.25. Find the value of x. 5

OR

A bag contains 4 red and 8 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. Find the value of x. 5

27. Water from a full cylindrical tank of radius 7 cm and height 12 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

OR

The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 2, 17, p, 10, 9 respectively is 66.4. Find the value of p. 5

END OF QUESTION PAPER — Total: 50 Marks

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